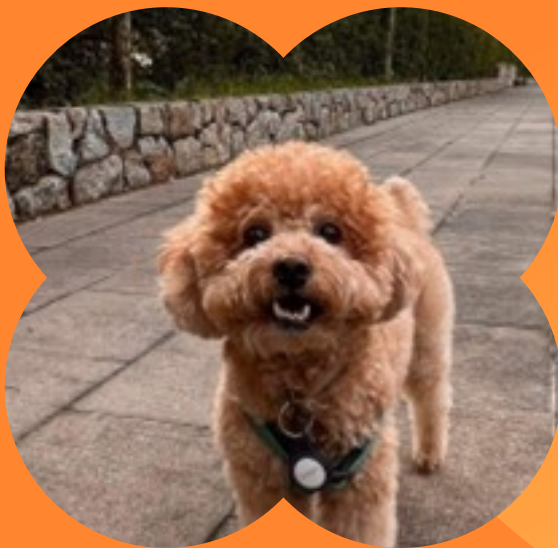


Midterm Report

"FurBot"

A web-based chatbot offering quick and accurate assistance for dog owners' everyday needs, from parks to grooming and supplies.



Demo Link:

https://youtu.be/_xS3aaV1EFw

Presented To

PADMAPRIYA ARASANIPALAI KANDHADAI

Presented By

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Introduction

I. Domain, Background, and Context

FurBot was inspired by a personal journey as a dog owner migrating to Canada, facing challenges in navigating an unfamiliar environment. Without established local connections, finding dog-friendly facilities, reliable grooming services, and essential pet care resources was overwhelming. This experience highlighted a significant gap in accessible, localized, and reliable pet care information, particularly for newcomers and urban pet owners.

To bridge this gap, FurBot was developed as a web-based chatbot designed to provide pet owners with quick and reliable access to essential services and information. FurBot uses predefined intents and responses alongside natural language processing (NLP) to accurately handle user queries. Whether locating nearby veterinary clinics, grooming salons, or dog parks, FurBot streamlines the search process, offering real-time recommendations tailored to users' locations.

Beyond information retrieval, FurBot now includes Pet Profile Management and Automated Reminders, allowing users to:

- *Create and manage pet profiles (store breed, age, and medical history).*
- *Set reminders for vet visits, vaccinations, and grooming appointments.*
- *Receive automated email notifications for upcoming pet care tasks, ensuring pets receive timely care.*

By consolidating essential pet care services into a single, interactive platform, FurBot enhances the efficiency, accessibility, and usability of pet-related resources, empowering pet owners with real-time assistance and a personalized experience.

II. Problem Statement

Key Questions Addressed:

- How can a chatbot provide reliable and real-time assistance to dog owners in finding localized pet care resources?
- What solutions can effectively bridge the gap between dog owners and small pet-related businesses?
- *How can a reminder system with email notifications improve pet care routines and reduce missed appointments?*

Pet owners frequently struggle to access reliable, localized information about pet-friendly facilities and services such as veterinary clinics, grooming salons, and dog parks. This challenge is further complicated when navigating process-related tasks like traveling with pets or understanding pet care regulations. *Additionally, many pet owners forget essential tasks like vaccinations and vet appointments, leading to gaps in their pet's healthcare.*

A centralized, chatbot-driven platform like FurBot, powered by NLP and automated reminders, can address these issues. By integrating predefined intents, real-time recommendations, and automated email reminders, FurBot helps pet owners stay informed, organized, and proactive about their pet's well-being.



III. Relevant Literature and Research

The Need for Localized Pet Care Services

During the pandemic, much of the focus shifted to human healthcare, often overlooking pet care accessibility. This created gaps in localized pet services, making it difficult for pet owners to find nearby veterinary clinics, grooming salons, and pet supply stores. FurBot addresses this gap by consolidating critical pet care information into a single, chatbot-driven platform.

Advancements in Chatbot Technology for Pet Owners

Recent innovations in chatbot technology have enhanced accessibility and convenience for pet owners:

- Woof.AI offers personalized product recommendations to improve pet care decisions (Woof.AI, 2024).
- AI-powered veterinary chatbots assist with scheduling, FAQs, and reminders, providing round-the-clock pet care support (FastBots, 2024).

Building on these advancements, FurBot integrates NLP with an automated reminder system, ensuring real-time assistance and proactive pet care. By delivering location-specific recommendations, personalized pet reminders, and email notifications, FurBot simplifies access to essential services while keeping pet owners informed and engaged.



Pet-friendly Parks



Pet Stores



Pet Salon



Pet Hospital

IV. Hypothesis, Assumptions, and Potential Benefit of Research

Hypotheses and Assumptions

FurBot aims to address three key challenges:

1. Access to reliable pet care resources – Localized recommendations will reduce search time for essential services.
2. *Personalized pet care management – Automated reminders will help pet owners stay on schedule for vaccinations, vet visits, and grooming.*
3. User engagement through real-time assistance – An interactive Q&A chatbot will provide instant responses, improving usability and satisfaction.

Key Assumptions:

- *Pet owners prefer a centralized platform for quick access to multiple pet-related services.*
- Personalized and location-specific recommendations enhance user experience.
- Real-time assistance via a chatbot is more practical and engaging than traditional search methods.

Expected Benefits

For Pet Owners:

- Time-saving and efficient – Quickly find pet-friendly resources without lengthy searches.
- *Personalized pet management – Stay organized with a centralized pet profile and automated reminders.*
- *Proactive pet care – Receive timely email alerts to ensure important pet care tasks are completed on schedule.*

For Local Businesses & Communities:

- Increased visibility – Local pet-related businesses gain exposure through FurBot's recommendations.
- Better pet welfare – Improved access to vet clinics, pet-friendly spaces, and grooming services.
- Stronger community engagement – A shared platform fosters connections between pet owners and local businesses.





Summary of the Initially Proposed Research Project



I. Research Design and Objectives

Design

Develop a web-based chatbot that consolidates pet-related services into a single platform, offering localized recommendations, personalized suggestions, and a predefined interactive Q&A system. By leveraging natural language processing (NLP) and a dataset of frequently asked questions, FurBot aims to improve accessibility, user satisfaction, and engagement.

Objectives

1. Database Creation: Develop a comprehensive database to provide localized recommendations for pet-related services, including veterinary clinics, grooming salons, and dog parks.
2. User Profile Integration: Implement a user profile system that enables personalized recommendations based on pet owners' preferences and needs, aligned with their location and service requirements.
3. Interactive Q&A System: Design a predefined question-and-answer system that delivers real-time assistance for common pet care inquiries, supported by structured datasets to ensure accuracy and relevance.

II. Methodology

Data Collection

- Gather data from publicly available directories, including information on veterinary clinics, pet grooming salons, and dog parks.
- Research blogs, forums, and articles detailing personal experiences of pet owners, such as traveling with pets, finding pet-friendly accommodations, or accessing pet care services in new locations (e.g., "Bringing your pet to the US for a quick vacation").

Data Analysis

Analyze user behavior and preferences to refine recommendations. Predefined Q&A content will be structured based on the most frequently asked questions by pet owners.

Justification

A consolidated platform provides efficiency and convenience, addressing a common gap in existing tools. The use of a chatbot ensures real-time interaction and tailored assistance, distinguishing FurBot from traditional search methods.



Summary of the Initially Proposed Research Project



III. Technologies

Operating System

- Windows 11: The development environment is based on Windows 11.

Programming Languages/Frameworks

- Backend: Python will be used for the chatbot's core functionality, particularly for NLP-based intent recognition and entity extraction. If time permits, OpenAI's pre-trained models will be integrated to enhance conversational capabilities and overall user experience.
- Frontend: The user interface will be developed using HTML, CSS, and JavaScript. Flask will serve as the web framework to connect the backend with the frontend, ensuring seamless integration and functionality.

Database

- SQLite: A lightweight and efficient database will be used to store essential data such as user profiles, localized pet service information, and chatbot interactions.

Development Tools and Methodologies

- Version Control with GitHub: GitHub will be used for version control and repository management.
- Agile Methodologies with JIRA: JIRA will be utilized for sprint planning, tracking progress, and managing tasks to ensure iterative development and timely delivery of project milestones.

IV. Expected Results

- Accurate and personalized recommendations for pet-related services, such as veterinary clinics, grooming salons, and dog parks, tailored to user profiles and preferences.
- Practical applications, including a user-friendly chatbot that provides quick, reliable answers to frequently asked pet care questions, making it a valuable tool for pet owners.
- Enhanced user experience through the consolidation of services and real-time assistance, saving time and effort for users while improving access to essential pet care resources.



Changes to the proposal



I. Feature Expansion: Pet Profile Management

Change:

FurBot now allows users to create and manage pet profiles, storing essential details like breed, age, medical history, and upcoming vaccinations.

Justification:

- **Personal Experience:** Pet owners often forget important pet care tasks, such as vaccinations and vet check-ups. A personalized profile system allows FurBot to track and store pet details, making information easily accessible.
- **Improved Engagement:** Users now interact with FurBot beyond simple queries, making it a comprehensive pet management tool rather than just a chatbot.

II. Feature Addition: Reminders for Vet Visits & Grooming

Change:

FurBot now includes a reminder system where users can set alerts for vet visits, vaccinations, and grooming appointments.

Justification:

- **User-Centric Approach:** Interviews with pet owners showed that many forget vaccination due dates or annual vet check-ups. A built-in reminder system reduces missed appointments.
- **Competitive Analysis:** Other pet platforms like Woof.AI focus on product recommendations, but few offer an integrated reminder system—making FurBot stand out in the market.
- **Technical Feasibility:** Reminders can be efficiently implemented using CRON (for task scheduling) and MongoDB to store pet records.



Proposed Research Project



III. Feature Enhancement: Automated Email Notifications for Pet Care Reminders

Change:

FurBot now sends automated email reminders to users about upcoming vet visits, vaccinations, and grooming schedules.

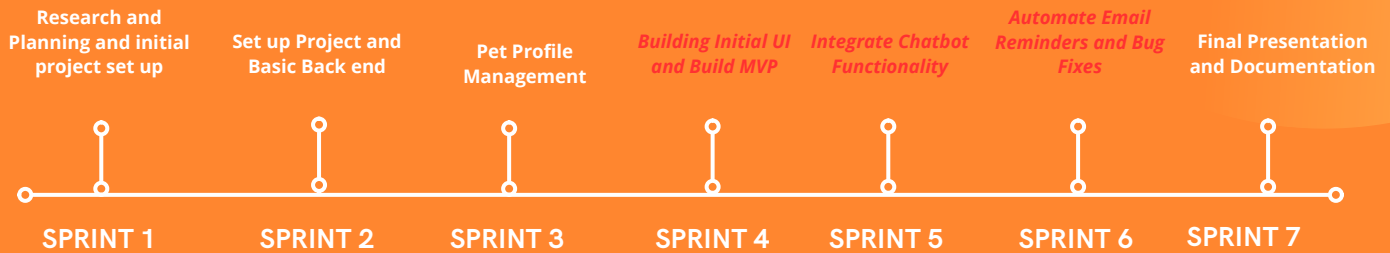
Justification:

- User Need: Many users may miss in-app notifications, but email reminders ensure they get alerts without opening the app.
- Improved Usability: An email system makes FurBot a proactive tool rather than just a passive chatbot.
- Technical Decision: Flask-Mail API was integrated into the backend for email automation, ensuring scheduled reminders are sent efficiently.

IV. Summary of the Changes

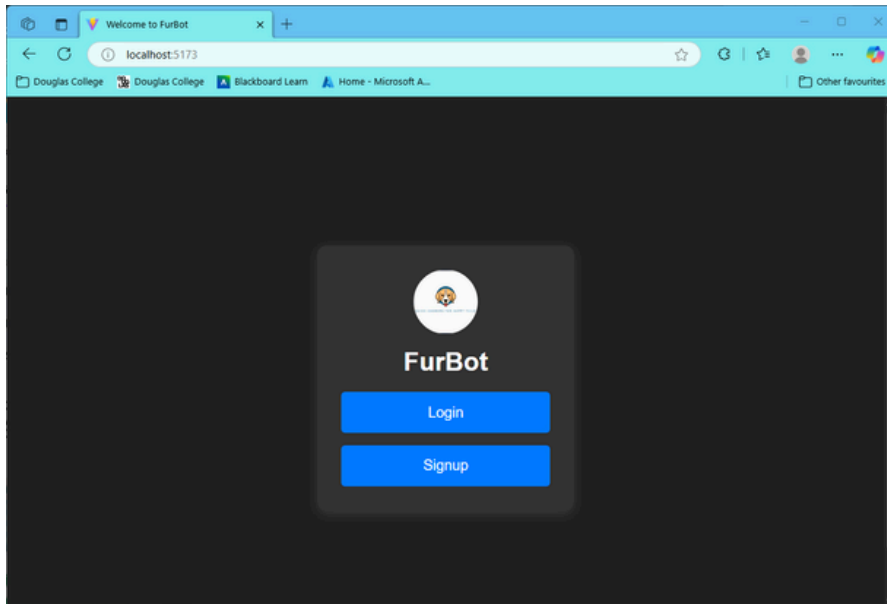
Category	Initial Proposal	Updated Proposal	Justification
Core Features	Basic chatbot answering pet-related queries	Added Pet Profile Management, Reminders, and Automated Email Notifications	Enhances usability by providing personalized assistance rather than just general information
Technology Stack	Python (Flask), NLP, MongoDB, Vue.js	Same tech stack with additional reminder scheduling via background jobs (Cron Jobs) and email notifications (Flask-Mail, SendGrid)	Required to implement automated reminders efficiently
Proposed Approach	Rule-based chatbot with predefined intents	Expanded chatbot functionality to include interactive pet care management	Provides a more engaging and practical tool for pet owners
Platform Scope	Web-based chatbot	Same (but backend enhanced to store pet data & automate reminders)	Expands FurBot's usefulness beyond real-time Q&A
Timelines	Focused on chatbot integration	Based on my initial Sprint Planning, I will still have the same number of Sprints with just few changes on the scope on each remaining sprints.	Ensures feature completeness before moving to advanced AI capabilities

Updated Project Planning and Timeline

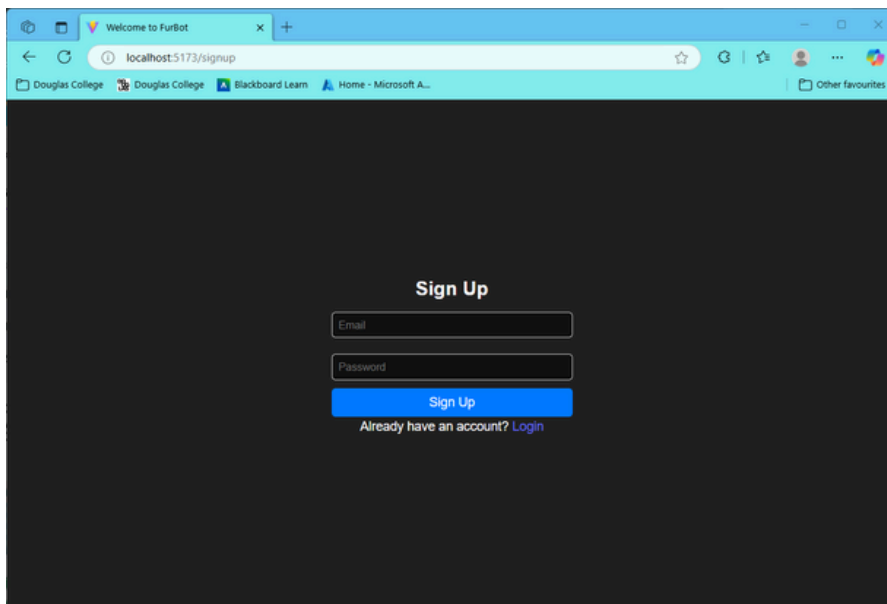


Status	Sprint	Weeks	Milestone
✓	Sprint 1	1 week	Complete research and planning, define project scope, objectives, and requirements. Set up project tools (e.g., GitHub, JIRA).
✓	Sprint 2	1 week	Set up project folder for implementation, basic backend set up (Database MongoDB and User Authentication)
✓	Sprint 3	2 weeks	Backend Setup: API Implementation for Pet Profile Management and Reminders.
✓	Sprint 4	2 weeks	Build initial UI and MVP: Login, Sign up, Adding Pets, Reminders and Progress Report #1.
⚙️	Sprint 5	2 weeks	Integrate Chatbot Functionality
	Sprint 6	1 week	Automate Email Reminders
	Sprint 7	1 week	Finalize end-to-end product

Implemented Features

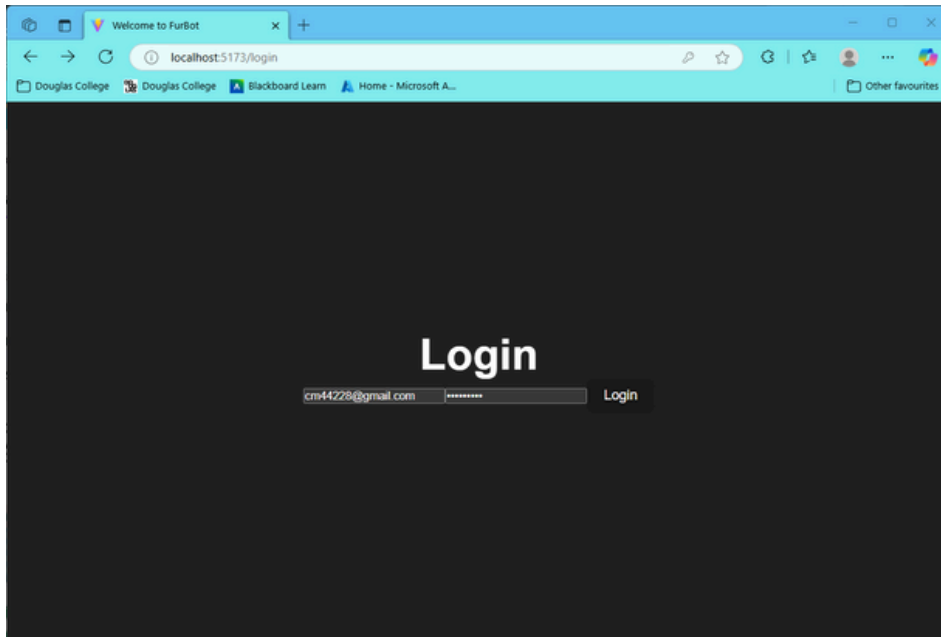


This is the homepage or the landing page of FurBot. It contains the Login for the registered users and a Signup option for new users.

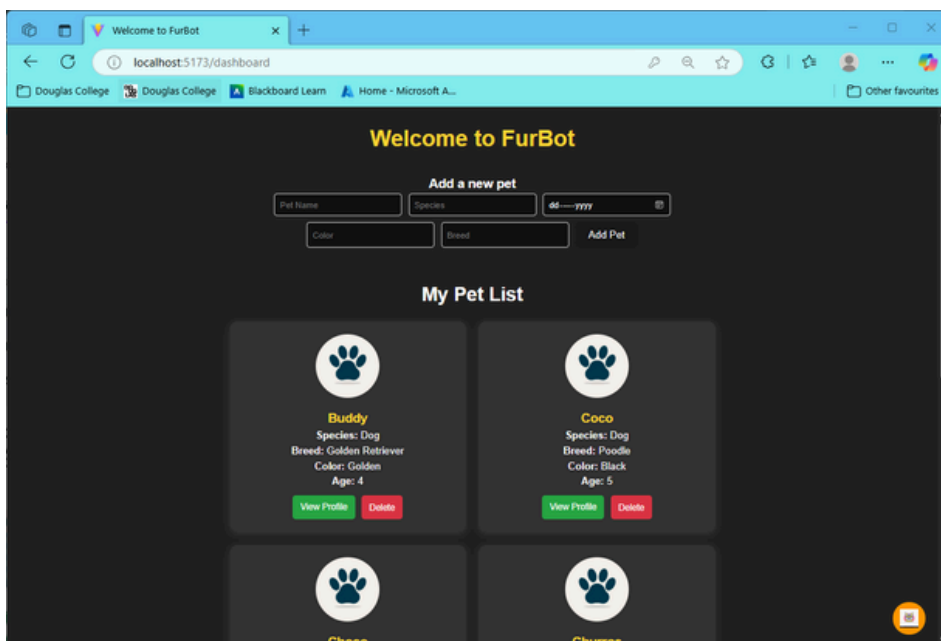


A routing to Sign Up page when user selected Signup from the homepage.

Implemented Features

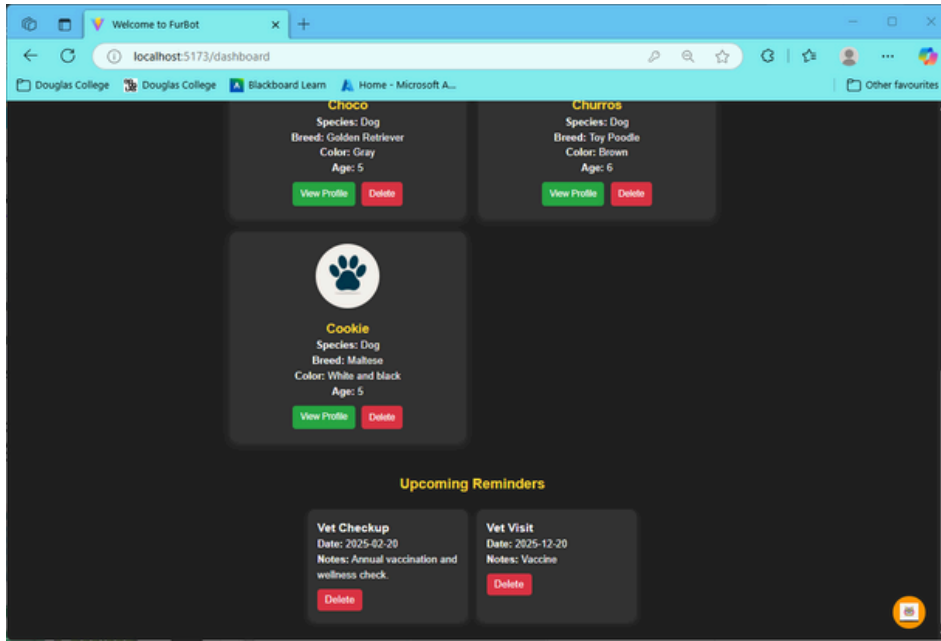


A routing to login page for the returning users and it will ask for email address and password.

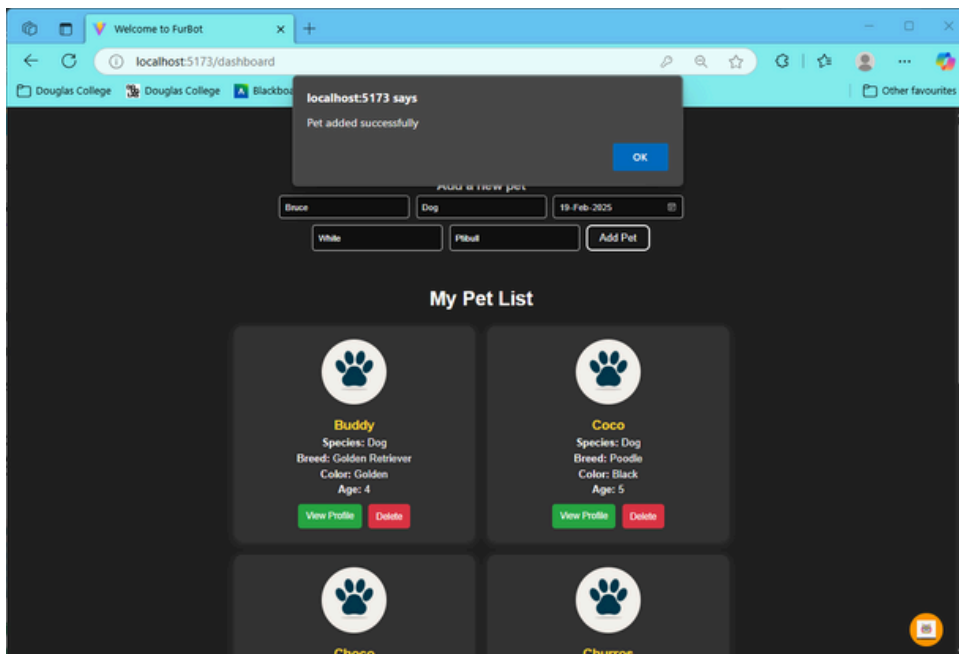


An authenticated user will be routed to it's Pet Dashboard which contains the pet list and form to add a pet.

Implemented Features

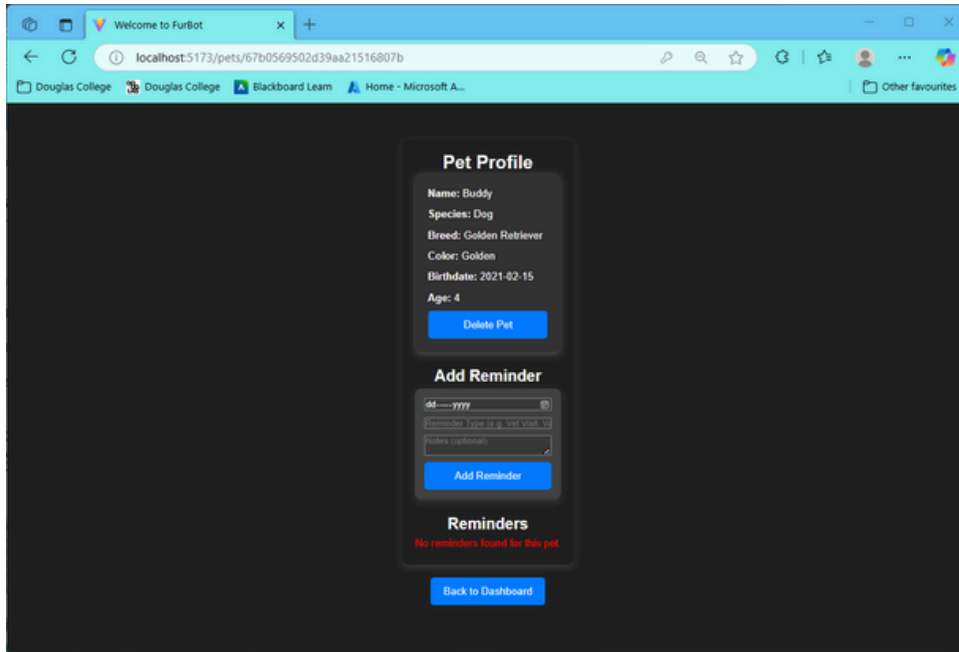


Pet Dashboard also let the authenticated user to view the Upcoming Reminders.

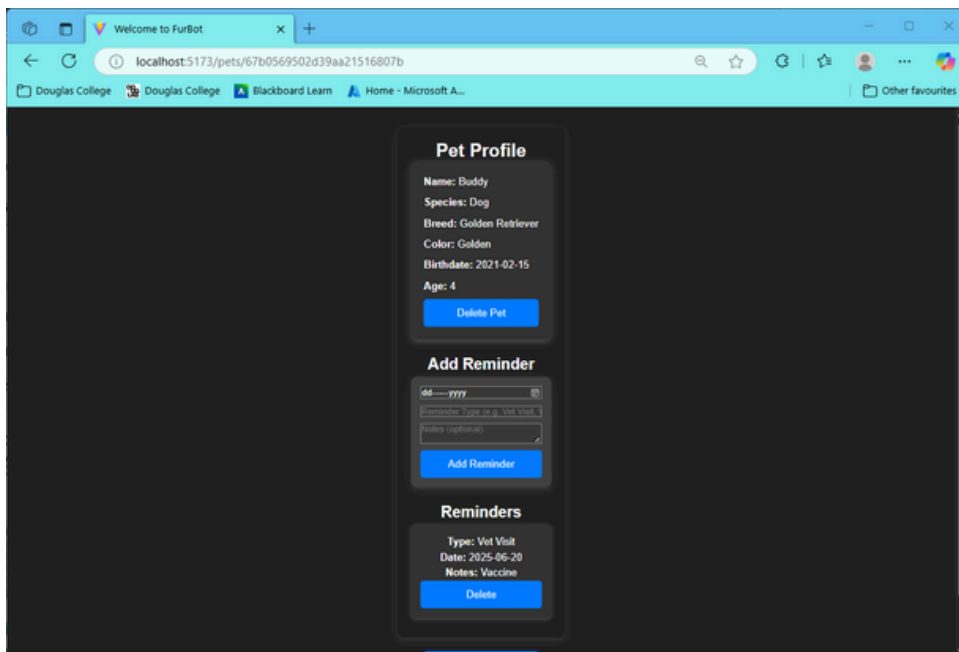


Alert will pop up once the pet is added successfully.

Implemented Features

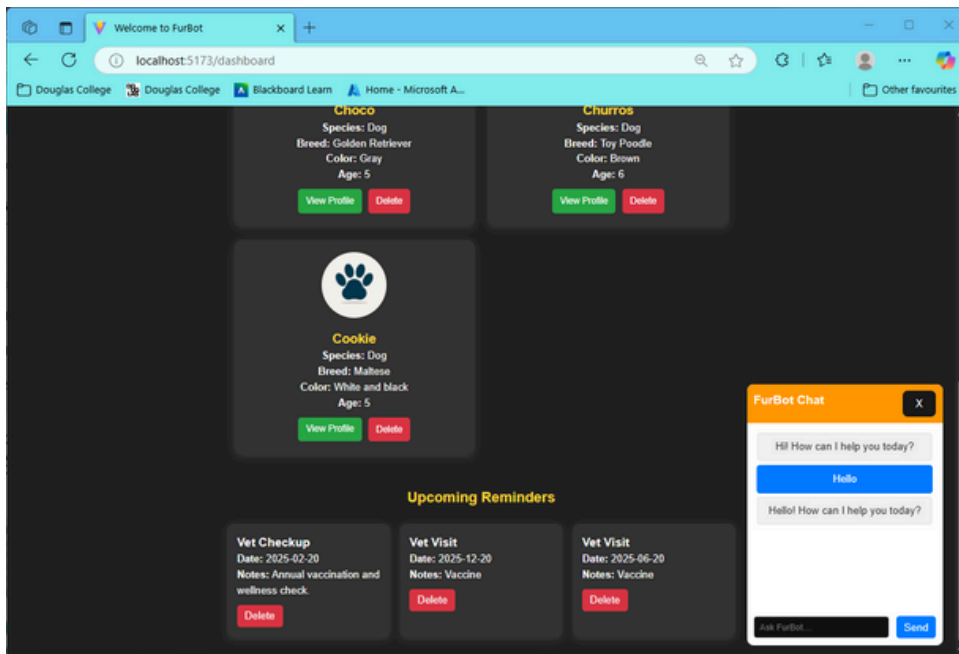


User can click the pet card to view the pet profile page.



User can add a reminder in each pet profile page.

Implemented Features



This is the chatbot UI, however it only provides the basic intents currently.

Work Date/Hours Logs



Date Start Working	Issue key	Summary	Hours Spent	Status	Sprint
19-02-2025	SCRUM-26	Create API Endpoint for Chatbot (/chatbot) and Simple UI Integration	5	Done	Sprint 5 - Chatbot Integration
18-02-2025	SCRUM-25	UI testing & debugging	8	Done	Sprint 4 - Building Initial UI
18-02-2025	SCRUM-23	API integration for authentication & pet management	3	Done	Sprint 4 - Building Initial UI
17-02-2025	SCRUM-24	User session handling (store token & authentication state)	4	Done	Sprint 4 - Building Initial UI
17-02-2025	SCRUM-22	Basic UI for Adding & Viewing Pets	2	Done	Sprint 4 - Building Initial UI
17-02-2025	SCRUM-21	Functional UI for Login & Sign-Up	2	Done	Sprint 4 - Building Initial UI
11-02-2025	SCRUM-20	Set up Job Scheduler to automate the email reminders	3	Done	Sprint 3 - Backend Setup
09-02-2025	SCRUM-17	Set up Reminders	3	Done	Sprint 3 - Backend Setup
09-02-2025	SCRUM-16	Create end points for pet profiles	2	Done	Sprint 3 - Backend Setup
09-02-2025	SCRUM-15	Set up MongoDB for Pet Profiles	1	Done	Sprint 3 - Backend Setup
06-02-2025	SCRUM-18	Progress Report #1	2	Done	Sprint 4 - Building Initial UI
06-02-2025	SCRUM-14	Test API	1	Done	Sprint 2 - Set Up the Backend
06-02-2025	SCRUM-12	Build User Authentication API	3	Done	Sprint 2 - Set Up the Backend
05-02-2025	SCRUM-11	Set Up MongoDB	1	Done	Sprint 2 - Set Up the Backend
05-02-2025	SCRUM-10	Install Dependencies	1	Done	Sprint 2 - Set Up the Backend
05-02-2025	SCRUM-9	Set Up Python and Virtual Environment	1	Done	Sprint 2 - Set Up the Backend
05-02-2025	SCRUM-8	Create the Folder Structure in the Implementation folder	1	Done	Sprint 2 - Set Up the Backend
25-01-2025	SCRUM-1	Draft Project Proposal	6	Done	Sprint 1 - FurBot Setup
19-01-2025	SCRUM-6	Set Up Project Repository	1	Done	Sprint 1 - FurBot Setup
19-01-2025	SCRUM-2	Set Up JIRA for Project Management	2	Done	Sprint 1 - FurBot Setup
18-01-2025	SCRUM-13	Research a topic for the project proposal	2	Done	Sprint 1 - FurBot Setup
18-01-2025	SCRUM-7	Research and Identify Technologies for FurBot	2	Done	Sprint 1 - FurBot Setup



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- Veterinary Chatbots - FastBots. (2024). Enhancing pet care with AI: The role of veterinary chatbots. Retrieved from <https://www.fastbots.ai/blog>
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Appendix



- **GitHub Repo** - https://github.com/charlz1202/W25_4495_S1_CharlieM
- **JIRA Project** - <https://charliemedialdea.atlassian.net/jira/software/projects/SCRUM/summary>
- **MVP Demo** - https://youtu.be/_xS3aaV1EFw